3.5.1 Energy, energy transfers and energy resources

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 1			
Describe, for common situations, the changes involved in the way energy is stored when a system changes. For example: • an object projected upwards • a moving object hitting an obstacle • a vehicle slowing down • bringing water to a boil in an electric kettle. Students may be required to describe the intended energy changes and the main energy wastages that occur in a range of devices.	Energy storage is limited to kinetic, gravitational potential, thermal and elastic potential.	6.2.1.1	4.1.1.4, 4.7.1.9 and 4.7.2.3
Outcome 2			
Energy can be transferred usefully, stored or dissipated, but cannot be created or destroyed.		6.1.2.1	4.8.2.5

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
The idea of efficiency. Whenever there are energy transfers in a system only part of the energy is usefully transferred. The rest of the energy is dissipated so that it is stored in less useful ways. This energy is often described as being 'wasted'. Unwanted energy transfers can be reduced in a number of ways, eg through lubrication and the use of thermal insulation.	Calculations of efficiency will not be required. Examples of the application of the idea of efficiency could include domestic appliances and vehicles.		4.8.2.6
How the rate of cooling of a building is affected by the thickness and thermal conductivity of its walls. The higher the thermal conductivity of a material the higher the rate of energy transfer by conduction across the material.	**Suggested activity for TDA** Investigate factors that affect the rate of cooling of a container of water, eg surface area, initial temperature, types of insulation, colour of the container. **Suggested activity for TDA** Investigate the thermal conductivity of different materials, eg which is better for a saucepan handle: wood or metal? Students do not need to know the definition of thermal conductivity.	6.1.2.1	4.8.2.6
Outcome 3			
Describe the main energy resources available for use on Earth. These include fossil fuels (coal, oil and gas), nuclear fuel, bio-fuel, wind, hydro-electricity, geothermal, the tides, the Sun, water waves. Distinguish between energy resources that are renewable and energy resources that are non-renewable.	Descriptions of how electricity is generated in a power station are not required other than the idea that a turbine is used to turn a generator.	6.1.3	4.8.2.4